
Probe Invariance in Code Models Is Largely Lexical

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Abstract

1 A linear probe on a code model’s residual stream recovers a variable’s *role* almost as
2 well in a language it was not fitted on as in the one it was: across six ordered pairs of
3 Python, JavaScript and PHP, matched problem by problem, transfer moves -0.012
4 macro-F1 at a boundary where a model-free character n -gram classifier loses 0.126.
5 Read at face value that says role representations are language-independent. It is
6 largely an artifact. The probe pools activations over the occurrence’s own tokens, so
7 it reads the variable’s name. The baseline is forbidden to. Pooling only surrounding
8 tokens, with the forward pass unchanged, costs the probe 65% of its margin over
9 an untrained network of the same architecture and pooling ($+0.269 \rightarrow +0.093$)
10 and multiplies its boundary effect fivefold ($-0.012 \rightarrow -0.061$), leaving it no
11 flatter than the untrained network itself (-0.042). The invariance was mostly the
12 identifier. We report this as a caution for cross-lingual probing of code models,
13 together with the model-free comparator such studies have lacked.

14 1 Introduction

15 Probes are how we ask what a code model represents, and cross-lingual probes ask whether what it
16 represents is tied to the language it saw. Prior work has fitted probes on one programming language
17 and evaluated on another [Hernández López et al., 2026], found language-specific components that
18 can be removed from code embeddings [Utpala et al., 2024], and shown that alignment between
19 languages on parallel programs is uneven [Kargaran et al., 2025]. That literature has an untrained-
20 network floor but no model-free one: no non-neural classifier is fitted on the same items, so a probe’s
21 transfer curve has nothing to be read against except itself.

22 We supply that comparator, and it changes the reading. A *variable role* [Sajaniemi, 2002], such as
23 accumulator or index key, is a property of what a variable does rather than of its name or its language,
24 so it is a case where a language-independent representation is plausible on its face. Fitting a role
25 classifier on one language and testing on another, the probe looks strikingly invariant beside a surface
26 baseline that is not.

27 The comparison is unfair in a way that turns out to decide the result. The probe pools activations over
28 the occurrence’s own tokens and therefore reads the identifier, which the surface baseline has masked
29 out. Removing that asymmetry removes most of the effect. What survives is a smaller finding about
30 the surface and a methodological caution about the probe.

31 2 Setup

32 **Corpus.** XLCOST [Zhu et al., 2022] provides parallel solutions keyed by a problem identifier stable
33 across languages. Only Python, JavaScript and PHP share problems at usable rates. No other pair
34 in XLCOST reaches 500 shared problems, including same-family pairs such as C# and Java, which
35 share 175 (Appendix A). We restrict all three languages to the 869 problems all three share, so every

Table 1: Cross-lingual role transfer, macro-F1 averaged over three roles. *Close* is the two directions between JavaScript and PHP. *Python* is the four transfers where Python is one of the two languages. Each probe row is paired with an untrained network at its own pooling.

	close	Python	effect
Surface n -gram, name masked (no model)	0.929	0.804	−0.126
Probe, span-pooled (reads the name)	0.852	0.840	−0.012
untrained, span-pooled	0.590	0.568	−0.022
Probe, context-pooled (no name)	0.630	0.569	−0.061
untrained, context-pooled	0.524	0.482	−0.042

ordered pair scores identical problem ids and a difference between cells cannot be a difference in subject matter. Roles are extracted per occurrence and split by a frozen hash of the problem id.

The four conditions. All are scored on the same occurrences.

- **Surface n -gram.** A character n -gram classifier over the context around an occurrence with every instance of the variable’s name replaced by a placeholder, fitted on the source language and applied to the target. No model. We fix one masked-context variant in advance rather than taking the best per cell. The per-cell maximum is selected on the evaluation fold, and every fixed variant gives a *larger* boundary effect than it.
- **Probe, span-pooled.** Logistic regression on residual activations mean-pooled over the occurrence’s own tokens, layer chosen on source-language validation data. This reads the identifier.
- **Probe, context-pooled.** The same, pooling the 16 tokens either side of the occurrence and *excluding* its own. The forward pass is byte-identical, and only what is read changes. Masking the name in the source would also change the input distribution.
- **Untrained.** The same architecture and tokenizer with randomly initialised weights, run at both poolings. A probe score means little without a bound on what architecture and labels alone supply [Hewitt and Liang, 2019, Zhang and Bowman, 2018, Belinkov, 2022], and a context-pooled probe must be read against a context-pooled floor: the two poolings read different token sets, so their achievable scores differ for reasons unrelated to training.

Models are Qwen2.5-Coder-1.5B [Hui et al., 2024] and, for the untrained control, the architecture of the base model it is continued from [Qwen et al., 2024]. Confidence intervals resample whole problems, never occurrences, since occurrences inside one program share a forward pass.

3 The surface baseline depends on the language

The n -gram baseline reaches 0.929 between JavaScript and PHP without a model and without seeing the variable’s name, and loses 0.126 on transfers to or from Python: 95% CI $[-0.160, -0.092]$, resampling the 427 test problems. The effect is not uniform across roles and is concentrated in iterator, which alone gives -0.225 while accumulator and index_key together give -0.059 . Iterator is also the role whose extractors agree least across the boundary and whose intervals are twice as wide, so we treat the aggregate as an upper bound on how much of this is about language rather than labelling.

4 The probe looks invariant, and mostly should not

Read against that, the span-pooled probe is remarkable: it moves -0.012 , an order of magnitude less, and clears an untrained network of the same architecture by $+0.269$. That is the result the obvious reading calls language-independent role representation.

It does not survive removing the identifier. The probe pools over the occurrence’s own tokens, so the variable’s name is inside what it reads, while the baseline it is being compared against has that

72 name masked. Pooling only the surrounding tokens, with the forward pass unchanged, the probe falls
73 from 0.844 to 0.589 overall. Against its *matched* untrained floor of 0.496 that is a margin of +0.093,
74 against +0.269 before: 65% of what the probe knew came from the identifier.

75 **So did the invariance.** The boundary effect grows from -0.012 to -0.061 , and the untrained
76 network at the same pooling moves -0.042 . With the identifier removed the trained probe is no
77 flatter than an untrained one. The apparent language-independence was not a property of the role
78 representation. It was a property of variable names, which parallel implementations of the same
79 problem tend to share.

80 **What remains.** A margin of +0.093 over a matched floor is small but consistent in both groups,
81 so the model does encode something about roles beyond the identifier. We do not claim it is
82 language-independent: at that pooling its boundary effect is indistinguishable from an untrained
83 network's.

84 5 Discussion

85 The finding is methodological, and it generalises past this corpus. Any probe that pools over a span
86 containing an identifier can read that identifier, and in parallel corpora identifiers are among the most
87 language-invariant things present. A cross-lingual probing result obtained that way will overstate
88 representational invariance by however much the naming happens to agree, and no untrained-network
89 control detects this: the untrained network reads the same identifier and is defeated by the same
90 confound. Only a comparator that cannot see the name, or a pooling that excludes it, separates the
91 two.

92 That is also why the model-free arm matters. The surface baseline is the thing that made the
93 asymmetry visible, because it was the only condition with the identifier withheld. Prior cross-lingual
94 probing of code models [Hernández López et al., 2026, Utpala et al., 2024, Kargaran et al., 2025]
95 has untrained-network floors but no model-free comparator, and would not surface this. The concern
96 generalises the standard caution that a probe reports what is decodable rather than what the model
97 uses [Belinkov, 2022, Voita and Titov, 2020]: here part of what is decodable is the input itself, and it
98 survives the control usually relied on.

99 **Reconciling with prior alignment results.** Kargaran et al. [2025] find Python the *least* aligned of
100 these three languages on parallel programs, which sits oddly beside a span-pooled probe that barely
101 moves across the Python boundary. The two measure different things, whole-snippet retrieval against
102 role decodability, and our context-pooled result removes the tension: once the identifier is excluded,
103 our probe does show a boundary effect. What looked like disagreement was the confound.

104 **Limitations.** Three languages, of which one is the boundary, so the contrast has two points and
105 no gradient, and Appendix A shows XLCOST admits no wider version. One model at 1.5B. The
106 identifier-lift figures are aggregated over three roles and we do not report them per role. Iterator's
107 labels come from a different extractor path than Python's, its minority class falls to ten instances in
108 some cells, and it carries most of the surface effect. The accumulator and index_key figures are the
109 more trustworthy ones. Context pooling excludes the identifier but not everything correlated with it,
110 so +0.093 is an upper bound on identifier-free role information. Finally, this is a probing result: it
111 says what is decodable, not what the model uses.

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157 A Pairwise problem overlap in XLCoST

Table 2: Shared problem identifiers between every pair of XLCoST languages, train split. Matched cross-lingual transfer needs a pair to share enough problems to hold out a test fold; only the three bold cells clear 500. No pair outside Python/JavaScript/PHP does, including same-family pairs.

	C++	C#	Java	JavaScript	PHP	Python
C++	—	115	122	75	17	74
C#	115	—	175	75	14	62
Java	122	175	—	85	11	66
JavaScript	75	75	85	—	1,529	2,953
PHP	17	14	11	1,529	—	1,145
Python	74	62	66	2,953	1,145	—

158 B Full transfer table

Table 3: All eighteen transfer cells for Qwen2.5-Coder-1.5B. *Surface* is the strongest masked-context n -gram feature; *name* uses the variable’s name alone; *probe* is the residual-stream probe. Majority and shuffled-label controls sit near chance throughout, and ρ is the macro-F1 movement from one test occurrence changing its prediction.

role	pair	n	surface	name	probe	major.	shuf.	ρ
accumulator	JavaScript→PHP	1,976	0.951	0.897	0.904	0.361	0.470	0.0005
accumulator	JavaScript→Python	3,341	0.813	0.873	0.875	0.415	0.473	0.0004
accumulator	PHP→JavaScript	1,903	0.954	0.897	0.918	0.372	0.490	0.0005
accumulator	PHP→Python	1,377	0.782	0.824	0.833	0.276	0.439	0.0008
accumulator	Python→JavaScript	3,294	0.785	0.874	0.918	0.370	0.504	0.0003
accumulator	Python→PHP	1,478	0.736	0.845	0.874	0.271	0.466	0.0007
index_key	JavaScript→PHP	1,976	0.940	0.834	0.897	0.384	0.478	0.0005
index_key	JavaScript→Python	3,341	0.859	0.787	0.883	0.330	0.497	0.0003
index_key	PHP→JavaScript	1,903	0.964	0.838	0.880	0.376	0.516	0.0005
index_key	PHP→Python	1,377	0.839	0.782	0.854	0.387	0.515	0.0008
index_key	Python→JavaScript	3,294	0.871	0.822	0.915	0.314	0.470	0.0003
index_key	Python→PHP	1,478	0.858	0.838	0.874	0.419	0.459	0.0008
iterator	JavaScript→PHP	1,976	0.937	0.862	0.906	0.448	0.488	0.0008
iterator	JavaScript→Python	3,341	0.774	0.733	0.939	0.444	0.472	0.0005
iterator	PHP→JavaScript	1,903	0.965	0.873	0.851	0.446	0.478	0.0008
iterator	PHP→Python	1,377	0.576	0.788	0.865	0.428	0.462	0.0010
iterator	Python→JavaScript	3,294	0.790	0.769	0.918	0.466	0.484	0.0007
iterator	Python→PHP	1,478	0.741	0.788	0.902	0.475	0.438	0.0020

Table 4: Probe transfer per model, averaged within each group. The untrained network shares Qwen2.5-1.5B’s architecture and tokenizer with randomly initialised weights; it is the floor a representational claim must clear, and it is itself flat.

model	close pair	Python pairs	all	vs. untrained
Qwen2.5-1.5B	0.899	0.888	0.892	+0.316
Qwen2.5-Coder-1.5B	0.893	0.887	0.889	+0.314
Qwen2.5-1.5B (untrained)	0.590	0.568	0.575	—

159 C Transfer mechanism

Table 5: Transfer mechanism for the iterator role. *Surviving* is the share of the source classifier’s discriminative mass realised in the target; *flip* is the share of that mass whose sign reverses. Surviving mass is uncorrelated with transfer; flipped mass predicts it almost exactly. Source-weighted variants are given for robustness.

pair	F1	surviving	flip	agree	flip (src-wt.)
PHP→JavaScript	0.965	0.735	0.072	+0.895	0.072
JavaScript→PHP	0.937	0.698	0.052	+0.945	0.068
Python→JavaScript	0.790	0.862	0.178	+0.695	0.211
JavaScript→Python	0.774	0.757	0.232	+0.811	0.241
Python→PHP	0.741	0.696	0.280	+0.528	0.367
PHP→Python	0.576	0.697	0.366	+0.476	0.346

160 D Language geometry

161 **The typing axis is not privileged.** Probing the same activations under every alternative 4-versus-3
162 partition of the seven languages ($\binom{7}{4} = 35$ groupings minus the real one, so 34 controls) gives

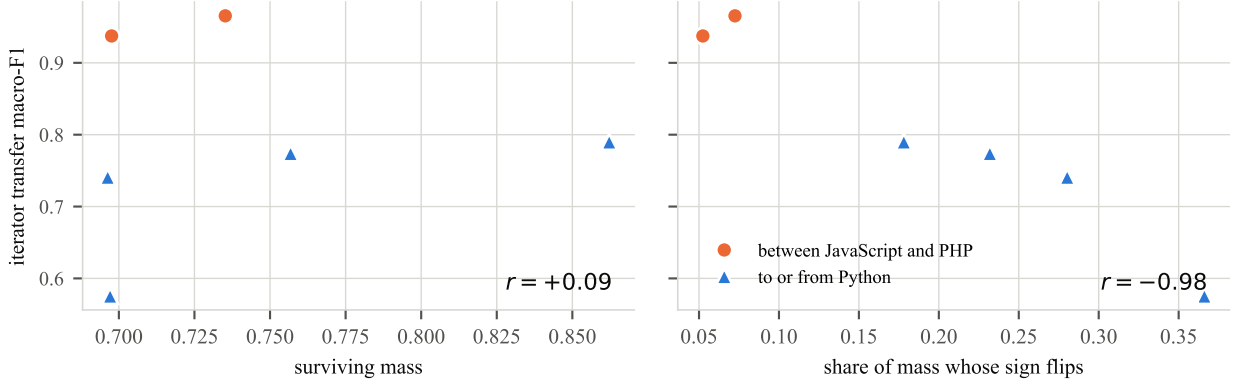


Figure 1: Two candidate explanations for the transfer gap, over the six ordered pairs. Left: how much of the source classifier’s discriminative mass the target realises, which is essentially unrelated to transfer. Right: the share of that mass whose sign reverses between a source-fitted and a target-fitted classifier, which tracks it closely. The cues are present in both groups; what changes is where they point.

Table 6: Masking an entire syntactic character class on both sides of the transfer. No class accounts for more than 0.021 of the 0.39 gap, so the realignment is distributed rather than carried by block delimiters or statement terminators.

pair	masked class	macro-F1	Δ
PHP→JavaScript	terminator	0.9496	-0.0159
PHP→JavaScript	brace	0.9645	-0.0010
PHP→JavaScript	bracket	0.9711	+0.0057
PHP→JavaScript	paren	0.9609	-0.0045
PHP→JavaScript	operator	0.9441	-0.0213
PHP→Python	terminator	0.5786	+0.0030
PHP→Python	brace	0.5750	-0.0006
PHP→Python	bracket	0.5733	-0.0023
PHP→Python	paren	0.5947	+0.0191
PHP→Python	operator	0.5801	+0.0045

best-layer macro-F1 0.9959 on average (min 0.9918, max 1.0000), against 1.0000 for the true static/dynamic split; 5 of the 34 controls match or exceed it. Any grouping of these languages is near-perfectly decodable, so the decodability of the typing split is evidence that language identity is encoded, not that type discipline is. At layer 4, where $|r|$ peaks at 0.941, the remaining components carry almost none of the label: PC2 $r = -0.014$, PC3 $r = +0.026$, PC4 $r = +0.171$.

E Causal interventions

Scope. Causal interventions were run for the *boolean* role over 5 languages (C++, Java, JavaScript, PHP, Python), 3 models, and 3 intervention modes (ablate, patch, steer), giving 375 layer-wise measurements. They are reported here rather than in the main text because they were run *within* languages and on a role outside the three this paper transfers, so they cannot support its central claim. Specificity is $|\text{clean} - \text{intervened}|$ divided by the same quantity for a random-position control: how much of the effect is specific to the variable’s own token positions rather than to editing anything at all. Two columns are given because the layer with the largest effect is generally not the layer with the best specificity, and quoting either alone misleads.

F Reproducibility

Every figure is regenerated from committed result files, and a regression test recomputes each of them from those files rather than quoting them. Activation extraction is the only step needing an

Table 7: Point-biserial correlation between PC1 of last-token residual activations and a static/dynamic typing label, by layer, with PC1’s explained-variance ratio. The sign of r is arbitrary (PCA fixes the axis only up to reflection, and each layer is refit independently); magnitude is what carries meaning.

layer	$ r $	EVR	layer	$ r $	EVR
0	0.883	0.481	15	0.894	0.309
1	0.898	0.447	16	0.882	0.303
2	0.914	0.391	17	0.874	0.296
3	0.939	0.373	18	0.873	0.289
4	0.941	0.370	19	0.873	0.299
5	0.862	0.369	20	0.903	0.291
6	0.828	0.361	21	0.886	0.276
7	0.844	0.342	22	0.859	0.265
8	0.896	0.339	23	0.842	0.252
9	0.919	0.333	24	0.812	0.242
10	0.880	0.327	25	0.774	0.253
11	0.893	0.302	26	0.694	0.255
12	0.917	0.292	27	0.732	0.237
13	0.904	0.314	28	0.576	0.242
14	0.898	0.309			

180 accelerator. The probe is logistic regression on activations mean-pooled over either the occurrence’s
181 span or the 16 tokens either side of it excluding the span; the surface baseline is a `char_wb` tf-idf
182 vectoriser with $n \in [2, 4]$ followed by logistic regression, fitted on the source language only. Pooling
183 and initialisation are recorded in each store’s identity, so a context-pooled store cannot be confused
184 with a span-pooled one downstream.

Table 8: Causal interventions on boolean-flag occurrences. *Peak-effect spec.* is specificity at the largest-effect layer; *best spec.* is the maximum over layers, with that layer named.

language	mode	model	peak-effect spec.	best spec. (layer)
C++	ablate	qwen2515b	4.5×	31.9× (L4)
C++	ablate	qwen25coder15b	4.4×	76.2× (L20)
C++	ablate	starcoder27b	4.3×	36.5× (L4)
C++	patch	qwen2515b	3.6×	7.9× (L20)
C++	patch	qwen25coder15b	3.6×	6.9× (L8)
C++	patch	starcoder27b	8.1×	10.4× (L16)
C++	steer	qwen2515b	45.0×	169.8× (L4)
C++	steer	qwen25coder15b	598.5×	598.5× (L24)
C++	steer	starcoder27b	6.9×	341.1× (L4)
Java	ablate	qwen2515b	28.5×	86.0× (L12)
Java	ablate	qwen25coder15b	12.3×	62.4× (L12)
Java	ablate	starcoder27b	27.0×	27.0× (L4)
Java	patch	qwen2515b	6.1×	7.7× (L12)
Java	patch	qwen25coder15b	5.9×	7.7× (L12)
Java	patch	starcoder27b	7.5×	8.6× (L12)
Java	steer	qwen2515b	72.1×	849.0× (L8)
Java	steer	qwen25coder15b	71.9×	186.6× (L0)
Java	steer	starcoder27b	252.0×	252.0× (L24)
JavaScript	ablate	qwen2515b	26.7×	26.7× (L16)
JavaScript	ablate	qwen25coder15b	28.1×	116.3× (L24)
JavaScript	ablate	starcoder27b	27.3×	27.3× (L4)
JavaScript	patch	qwen2515b	6.9×	6.9× (L8)
JavaScript	patch	qwen25coder15b	6.6×	7.4× (L12)
JavaScript	patch	starcoder27b	8.1×	8.2× (L16)
JavaScript	steer	qwen2515b	24.5×	76.4× (L8)
JavaScript	steer	qwen25coder15b	65.2×	149.5× (L4)
JavaScript	steer	starcoder27b	11.2×	60.8× (L28)
PHP	ablate	qwen2515b	15.0×	102.4× (L20)
PHP	ablate	qwen25coder15b	26.5×	79.2× (L20)
PHP	ablate	starcoder27b	19.4×	19.7× (L8)
PHP	patch	qwen2515b	7.2×	7.2× (L4)
PHP	patch	qwen25coder15b	4.7×	7.4× (L0)
PHP	patch	starcoder27b	6.3×	9.3× (L16)
PHP	steer	qwen2515b	190.3×	190.3× (L24)
PHP	steer	qwen25coder15b	11.5×	134.0× (L8)
PHP	steer	starcoder27b	8.7×	100.1× (L4)
Python	ablate	qwen2515b	5.8×	36.7× (L24)
Python	ablate	qwen25coder15b	6.8×	33.1× (L16)
Python	ablate	starcoder27b	26.5×	26.5× (L4)
Python	patch	qwen2515b	6.0×	7.2× (L16)
Python	patch	qwen25coder15b	6.7×	7.8× (L16)
Python	patch	starcoder27b	7.9×	9.7× (L12)
Python	steer	qwen2515b	209.1×	827.9× (L16)
Python	steer	qwen25coder15b	1025.8×	1025.8× (L24)
Python	steer	starcoder27b	8.6×	55.5× (L4)

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